GG-B-101G
29 April 1986
(SUPERSEDING)
GG-B-101F
17 November 1975

FEDERAL SPECIFICATION

BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED (Hard Coated)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hard-coated, plaster of paris bandages suitable for forming orthopedic casts.

1.2 Classification.

1.2.1 Types, classes, and sizes. Bandages shall be of the following types, classes, and sizes as specified (see 6.1):

Type I. - Fast Setting.

Class A - Rolled Bandages.
Sizes:
2 inches by 3 yards.
3 inches by 3 yards.
4 inches by 5 yards.
5 inches by 5 yards.
6 inches by 5 yards.
8 inches by 5 yards.

Class B - Splints.
Sizes:
3 inches by 15 inches.
4 inches by 15 inches.
5 inches by 30 inches.
5 inches by 45 inches.

AMSC N/A FSC 6510

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Type II. - Extra-Fast Setting.

Class A - Rolled Bandages.
Sizes:
2 inches by 3 yards.
3 inches by 3 yards.
4 inches by 5 yards.
5 inches by 5 yards.
6 inches by 5 yards.

Class B - Splints.
Sizes:
3 inches by 15 inches.
4 inches by 15 inches.
5 inches by 30 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

RR-S-366 - Sieve, Test.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of listed federal and military standards, specifications, and Commercial Item Description (CIDs) listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Military Specifications:

MIL-P-116 - Preservations, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Laws and Regulations:

Department of Health and Human Services (HHS):

Federal Food, Drug and Cosmetic Act and Regulations Promulgated thereunder.

(Application for copies should be addressed to the Division of Public Information, Food and Drug Administration, Washington, DC 20201).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

U.S. Pharmacopeial Convention, Inc.:

The Pharmacopeia of the United States of America (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials used in the manufacture of the bandages shall be of high quality and suitable in all respects for the purpose intended.

3.1.1 Gauze fabric. Fabric shall be either one of the following when tested in accordance with 4.4.3. Requirements for sterility are not applicable.

(a) Type II bleached absorbent gauze conforming to U.S. Pharmacopeia (U.S.P.) requirement.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.

(b) Commercial 28 by 20 (minimum) bleached gauze having a minimum weight of 23 grams per square yard and conforming to the U.S.P. absorbency requirement of not more than 30 seconds.

3.1.2 Plaster of paris. The plaster of paris used in the manufacture of the bandages shall be of high commercial quality and of good white color, suitable for fabrication of plaster of paris bandages. The addition of chemical modifiers such as binders, and stabilizing agents is permissible provided the end item meets all the requirements specified herein, and does not interfere with end use of bandage.

3.1.2.1 Fineness. A minimum of 99.0 percent of the plaster of paris shall pass through a No. 200 U. S. Standard sieve, when tested as specified in 4.4.4.

3.2 Design and construction. The bandages shall consist of a fabric impregnated with plaster of paris, and shall form a hard-coated cast and non-dusting surface. The plaster shall be uniformly spread and firmly bonded to the fabric. The finished bandages, when used in accordance with the manufacturer's instructions, shall be suitable for forming orthopedic casts for immobilization purposes. The finished bandages shall be white and shall be furnished as a neatly rolled bandage or a neatly folded splint, as applicable. The long edges of the rolled bandages and flat splints shall be pinked or furnished with a straight edge, free from raveled threads as classified in 4.3.

3.2.1 Class A, rolled bandages. Class A rolled bandages shall be furnished as a roll, in one continuous length of material. Each bandage shall be evenly rolled and shall have a center opening 1/4 to 7/8 inch (in.) in diameter, to provide rapid water penetration.

3.2.2 Class B, splints. Class B splints shall be furnishd flat in single layers of one continuous length of material, except for the 30 inch and 45 inch length which shall be folded at a 15 inch $\pm$ 1 inch interval.

3.3 Dimensions. Bandages shall conform to the dimensions listed in table I for the applicable class and nominal size.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the currentversion before use.

TABLE I. Dimensions.

(1) Class	(2) Nominal Size 1/	(3) Minimum Width	(4) Minimum Length
A	2.0 inches by 3.0 yards	1.8 inches	2.9 yards
A	3.0 inches by 3.0 yards	2.8 inches	2.9 yards
A	4.0 inches by 5.0 yards	3.8 inches	4.9 yards
A	5.0 inches by 5.0 yards	4.8 inches	4.9 yards
A	6.0 inches by 5.0 yards	5.8 inches	4.9 yards
A	8.0 inches by 5.0 yards	7.8 inches	4.9 yards
B	3.0 inches by 15.0 inches	2.8 inches	14.5 inches
B	4.0 inches by 15.0 inches	3.8 inches	14.5 inches
B	5.0 inches by 30.0 inches	4.8 inches	29.1 inches
B	5.0 inches by 45.0 inches	4.8 inches	43.5 inches

1/ Nominal size of bandage is the minimum average size of the applicable bandage when sampled and examined in accordance with 4.2.1 and 4.3. Width and length of individual bandages shall be no less than the values listed in columns 3 and 4 for the applicable class and nominal size.

3.4 Physical properties.

3.4.1 Dry weight and cast strength. The bandages shall meet the dry weight and cast strength of table II for the applicable bandage or splint size, when tested as specified in 4.4.5 and 4.4.7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE II. Dry weight and cast strength requirements.

1/ Class A and Class B (Rolled Bandage and Splint)	Dry Weight in grams		Cast strength in pounds	
			Minimum	
Sizes	Minimum	Maximum	Type I	Type II
2 inches by 3 yards	62	76	80	70
3 inches by 3 yards	93	115	140	105
4 inches by 5 yards	207	256	450	300
5 inches by 5 yards	259	301	550	375
6 inches by 5 yards	311	379	650	450
8 inches by 5 yards	414	482	800	--
3 inches by 15 inches	90	106	140	105
4 inches by 15 inches	207	256	450	300
5 inches by 30 inches	258	315	550	375
5 inches by 45 inches	259	315	550	

1/ Bandages and splints prior to testing, shall measure 180 inches (5 yard bandage or equivalent splint size) or 108 inches (3 yard bandage or equivalent splint size) ± 2% in length and nominal width ± 1/8 inch. For respective splint sizes, see paragraph 4.4.2.

3.4.2 Water penetration. Bandages shall show uniform and complete water penetration, with no visible dry spots or areas, when tested as specified in 4.4.9.

3.4.3 Wet weight loss. The loss in weight of plaster of paris for the bandages shall not exceed 10.0 percent, when tested as specified in 4.4.6.

3.4.4 Setting time.

3.4.4.1 Type I, fast setting. Type I material shall have a setting time of 4.00 to 7.00 minutes, when tested as specified in 4.4.8.

3.4.4.2 Type II, extra-fast setting. Type II material shall have a setting time of 2.00 to 4.00 minutes, when tested as specified in 4.4.8.

3.5 Irritation. The bandages shall be non-irritating to the normal skin, when used as directed. The bandages shall be free from harmful material, and shall have no objectionable odor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

3.6 Instructions for use. Instructions for use shall be furnished for bandages (both rolled bandages, see 3.6.1, and splints, see 3.6.2). The instructions shall be printed on the unit package or box, or as an alternate on a separate sheet of white bond or offset paper, measuring 8 ± 1/2 inches wide and 5 ± 1/4 inches high. The printing with illustrations shall show no offsetting, smearing, or bleeding and shall be clearly legible. Any additional information the manufacturer may desire to enclose, shall be printed on a separate sheet or in a pamphlet.

3.6.1 Class A, rolled bandages. Each unit package for Class A rolled bandages shall be furnished with instructions conforming to the data and illustrations shown in figure 1 for type I and type II, class A material. Illustrations and wordage in figure 1 are typical only and are not intended to restrict or preclude the use of other equivalent commercial instructions for dipping, squeezing, and applying the bandage.

3.6.2 Class B, splints. Each unit box for class B splints shall be furnished with the following instructions:

INSTRUCTIONS

1. Pass desired number of splints (usually 5 to 8) through water, in one multiple layer unit.

2. Hold at an angle briefly to drain off excess water.

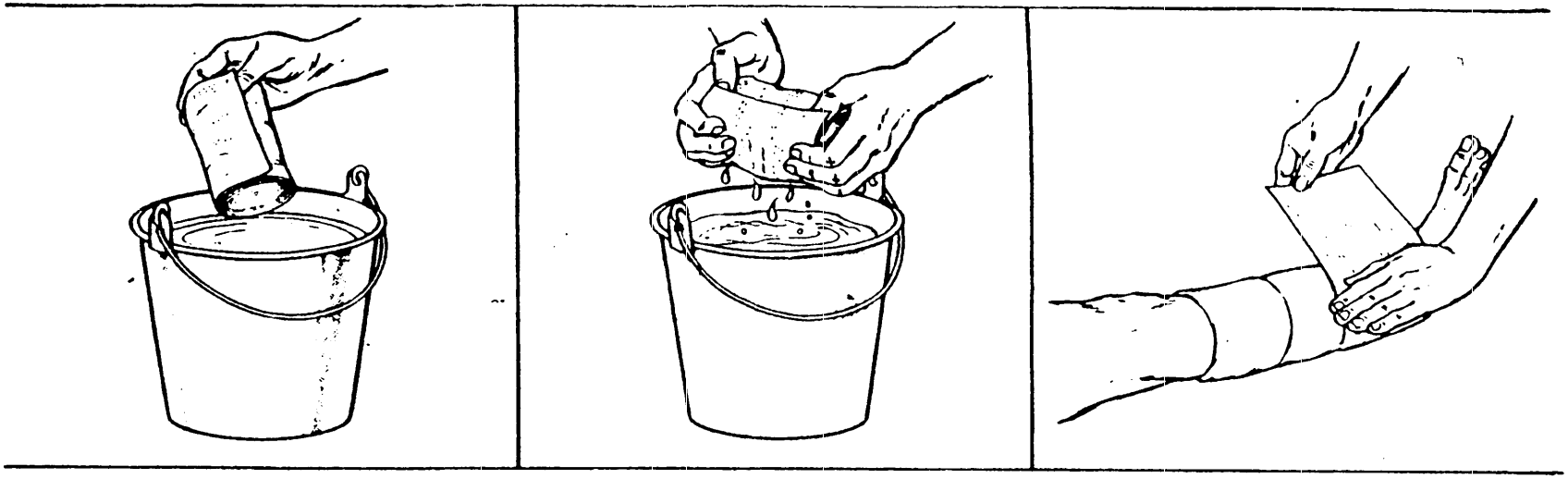
3. Press out lightly from center to each end to fuse splints into a solid slab, then apply."

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

exterior container. One hundred (100) intermediate packages shall

• INSTRUCTIONS FOR USE •

NOTE: DO NOT DIP MORE THAN ONE BANDAGE AT A TIME.



DIP

HOLD BANDAGE LOOSELY AND SUBMERGE IN CLEAN, TEPID WATER FOR 5 SECONDS (USE ALARM-TYPE TIMER). KEEP THUMB UNDER EDGE OF BANDAGE TO AVOID DIFFICULTY IN FINDING STARTING END AFTER SQUEEZING.

SQUEEZE

SQUEEZE BANDAGE WITH TWISTING MOTION TO EXPEL EXCESS WATER, KEEPING THUMBS UNDER STARTING END.

WATER IN PAIL MUST BE CLEAN AT ALL TIMES.

APPLY

IN APPLYING TO PATIENT, RUB EACH TURN OF BANDAGE INTO THE LAYERS BELOW TO FORM A SOLID, WELL-FUSED CAST. FAST-SETTING BANDAGES SET IN 4 TO 7 MINUTES. EXTRA-FAST SETTING BANDAGES SET IN 2 TO 4 MINUTES.

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T11:48Z
Check the source to verify that this is the current version before use.

3.7 Lot or batch number. Each unit container shall be marked with a lot or batch number.

3.8 Shelf life. The shelf life of the bandage shall be not less than 24 months from date of manufacture.

3.8.1 Delivery. Not more than four months shall have elapsed from the date of manufacture of the bandage to the date of delivery to the Government.

3.9 Workmanship. Bandages and splints shall be free of defects which may detract from their appearance or impair their serviceability. Bandages shall be furnished clean.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has ben maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the raw materials, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Manufacturer's protocol. A manufacturer's protocol, that lists the inspection characteristics with specification requirements and which states that inspection results comply with said requirements, is acceptable, providing the actual inspection results are made readily available to Government representatives upon request.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:48Z
Check the source to verify that this is the currentversion before use.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified in tables III, IV, V and VI (see 4.2.1, 4.2.2, 4.3 and 4.4.2) and 4.5.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as indicated in table III except for the minimum average dimensions. Unit of product for examination purposes shall be one bandage or one splint from each lot or batch number.

TABLE III. Inspection and acceptable quality levels.

Type defect	Inspection level	AQL (percent defective)
Major	S-2	1.5
Minor	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and table IV. Where feasible the same test unit may be used for determination of two or more test characteristics. Acceptance number shall be zero for all sample sizes. The unit of product for end item testing shall be one unit of issue from each lot or batch number.

Source: https://assist.dla.mil Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE IV. Testing Criteria.

(1) Component	(2) Characteristic	(3) Requirement	(4) Test procedure	(5) Minimum sample size	(6) Inspection level
Fabric	Gauze	3.1.1	4.4.3	--	1/
Plaster of Paris	Fineness	3.1.2.1	4.4.4	--	1/
Polyethylene bag	Leakage	5.1.1.1.3	4.4.10	--	S-1
End item:	Dry weight	3.4.1	4.4.5	5	S-2 2/
	Cast strength	3.4.1	4.4.7	5	S-2 2/
	Setting time	3.4.1	4.4.8	5	S-2 2/
	Water penetration	3.4.2	4.4.9	5 3/	S-2 2/
	Wet weight loss	3.4.3	4.4.6	2 4/	S-1 2/

1/ Sufficient quantity from each lot of raw material.
2/ One bandage or splint shall be tested from each unit of product (issue) taken in accordance with inspection level, provided no less than the minimum sample indicated in column (5) shall be taken where less is indicated in MIL-STD-105.
3/ Water penetration tested on same specimens used for setting time (see 4.4.9).
4/ Wet weight loss tested on same specimens used for dry weight (see 4.4.6.2).

4.3 Examination. The bandage shall be examined for defects, including, but not limited to, those listed in table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE V. Classification of Defects.

Categories	Defects
Major	
101	Edges not clean cut for a straight edge and not properly pinked for pinked edge.
102	Bandages not of type, class and size specified.
103	Center core not present (class A).
104	Surface not hard coated and not non-dusting.
105	Bandages not free of grease spots, dirt, or stains.
106	Raveling with free threads greater than 3 inches.
107	Objectionable odor present.
108	Plaster not uniformly spread on fabric.
109	Plaster not firmly bonded to fabric.
110	Bandages not of one continuous length.
Minor	
201	Bandages not neatly rolled (class A) or folded (class B).
202	Instructions not as specified.
203	Bandage not white in color.
204	Raveling with free threads greater than 2 and no more than 3 inches.
205	Width and length dimensions not as specified.
206	Center opening dimensions not as specified.

4.4 Tests.

4.4.1 Standard testing conditions. Testing shall be conducted at $72^{o} \pm 5^{o}$F ($22.2^{o} \pm 2.8^{o}$ Centigrade (C) and 50 ± 10 percent relative humidity, on bandages wrapped and ready for shipment by removing wrapping and testing within 15 minutes from removal of wrapping. Equipment and material shall be conditioned at this temperature and humidity at least ten hours prior to testing. For referee testing, tests shall be conducted under uniform atmospheric conditions of $70.0^{o} \pm 2^{o}$ F ($21.1^{o} \pm 1.1^{o}$ C) and 50 ± 10 percent relative humidity.

4.4.2 Specimen equivalency of splints. To determine dry weight, cast strength (see 3.4.1 and table II), wet weight loss and setting time on splints, specimens shall be prepared for testing by taking the required number of splints specified in table VI, and rolling them together. Each added splint shall be overlapped to form an external size diameter approximating the rolled equivalent bandage size with a center opening of 1/4 to 7/8 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

TABLE VI. Specimen equivalency of splints.

Splint Size	Number of splints to be used to make specimens	Equivalent Bandage Size
3 inches by 15 inches	7 plus enough to make 108 inches	3 inches by 3 yards
4 inches by 15 inches	12	4 inches by 5 yards
5 inches by 30 inches	6	5 inches by 5 yards
5 inches by 45 inches	4	5 inches by 5 yards

4.4.3 Fabric. Type II gauze shall be tested in accordance with the applicable test (except sterility) in the U.S.P. Commercial gauze shall be tested for absorbency in accordance with the U.S.P., and the weight shall be determined on a suitable balance.

4.4.4 Fineness. Fineness for plaster of paris is expressed as the value for the raw material prior to fabrication. Fineness shall be determined on a 100 gm sample of plaster of paris. The sample of plaster of paris shall be spread uniformly over the face of the screen of a U. S. Standard No. 200 sieve, conforming to the requirements of RR-S-366. Using a swirling motion on the screen, a soft stream of water at a temperature of $70.0^{\circ} \pm 2.0^{\circ}$ F shall be applied through a 5/8 inch inside diameter rubber hose, and the sample shall be rapidly washed through the screen. The washing operation shall be completed in 30 seconds. The sieve and retained sample shall be placed in $125.0^{\circ} \pm 5.0^{\circ}$ F oven and dried to a constant weight. The amount of material retained on the sieve shall be weighed and the fineness shall be calculated and expressed as a percentage of the weight of the original sample.

4.4.5 Dry weights. The dry weight of the intact bandage and splint specimen shall be weighed on a scale or balance having an accuracy of 0.1 gm. When bandage is defective in dry weight, bandages shall be checked for dimensions. If dimensions do not comply, bandage shall be considered defective in dimensions and not in dry weight. Another random sample shall be taken to replace the defective dimensional sample for retesting for dry weight compliance.

4.4.6 Wet weight loss.

4.4.6.1 Blank. The weight of a one-liter volumetric flask at a temperature of $25.0^{\circ} \pm 1.0^{\circ}$ C, filled with water at a temperature of $25.0^{\circ} \pm 1.0^{\circ}$C, shall be determined to the nearest 0.1 gm.

4.4.6.2 Procedure. Bandages and splints shall comply with the requirements for Dry Weight (table II) prior to testing for wet weight loss. The intact bandage or splint specimen (see 4.4.2) shall be weighed and then immersed in a dish of suitable size containing 900 cubic centimeters (cc) of water at a temperature of $25.0^{\circ} \pm 1.0^{\circ}$C. The test specimen shall be manipulated for 5 seconds in the water in such a manner that it is thoroughly wetted. The specimen shall then be removed, and the ends grasped with each hand, and squeezed without twisting for five seconds, with the expressed water being retained in the immersion dish. A spray bottle of water shall be used to wash off

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

any plaster residue from the hands into the immersion dish. The contents of the immersion dish shall then be transferred to the volumetric flask (see 4.4.6.1), upon which the blank has been run, by means of a funnel; care shall be exercised to remove all sediment from the immersion dish with additional water at $25.0^{o} \pm 1.0^{o}C$. The flask shall then be made up to volume and the weight determined to the nearest 0.1 gm. The difference in weight between the unknown and the blank shall then be determined. Conversion to weight of plaster of paris shall be determined by the following formula:

$$X = 1.5770Y - 1.2616$$

where

X = weight of plaster of paris (gm)

Y = unknown minus blank (gm)

The percentage plaster loss is calculated from:

$$\frac{\text{Weight of plaster}}{\text{Weight of bandage of splints}} \times 100$$

4.4.7 Cast strength.

4.4.7.1 Preparation of specimens. Bandages and splints shall comply with the requirements for Dry Weight (table II) prior to testing for Cast Strength. The intact rolled bandage or prepared splint specimen (see 4.4.2) shall be immersed vertically (with center opening vertical), in a liter beaker or other suitable receptacle of distilled water at a temperature of $25.0^{o} \pm 1.0^{o}$ C. There shall be a two to three inch head of water above the top of the test specimen. After five seconds of immersion, the test specimen shall be removed from the water and wrung out by hand until no further water can be expressed. Starting with the free edge, the specimen shall be wrapped around a wax paper lined, seamless copper or brass alloy tube or solid rod, 2 + 1/64 inches in diameter and concentric within .005 inch. The outside surface of the tube or rod shall have a bright smooth finish. In wrapping the specimen, extreme care shall be exercised to keep each subsequent layer coincident with the preceding layer. The specimen shall be wrapped and stroked with the hand to provide a firm bonding of the layers of the specimen. At the end of 20 minutes, measured from time of immersion of specimen, the hardened specimen shall be removed from the tube or rod.

4.4.7.2 Crushing procedure. At the end of one hour, measured from the time of immersion, the hardened (cast) specimen shall be crushed diametrically with the crushing jaws greater than the full length of the respective cast specimen. The longitudinal axis of the cast shall be at a right angle to the direction of travel of the compression tested. The cast specimen shall be crushed at a rate of one inch per minute until the external diameter is reduced to two inches. Test equipment used for determining cast strength shall be the Dillon Mfg. Co., Dynamometer; Testing Machines, Inc., Hinde and Dauche Crush Tester; or other suitable equipment, provided prior approval is received from the contracting officer.

4.4.8 Setting time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.4.8.1 Apparatus for setting time. Setting time shall be tested with the modified Vicat apparatus illustrated in figure 2, and as described in 4.4.8.2.

4.4.8.2 Procedure for setting time. The intact rolled bandage or splint specimen (see 4.4.2) shall be held at 22.2° ± 2.8°C for at least 10 hours before test. The intact rolled bandage or splint specimen shall be immersed vertically, (with center opening vertical) in a liter beaker of water at a temperature of 25.0° ± 1.0° C. There shall be a two to three inch head of water above the top of the test specimen. After five seconds of immersion, the test specimen shall be removed from the water and wrung out by hand until no further water can be expressed. The test specimen shall be wrapped convolutely about the setting time apparatus (see figure 2) which has been covered with one layer of waxed paper. Starting at the lower left hand side of the box, the first layer shall be passed over the top of the box. Each layer shall be laminated to the preceding layer and the wrapping shall continue convolutely until a pad five layers thick is built up. For the 3 inch by 3 yard bandage, upon completion of three-half convolutions, take the remaining free end and reverse the specimen back on itself, to form a five layer thickness in test area. In testing of splint specimen, free ends shall be sealed smoothly by overlapping sufficiently to form a firm specimen. Lapping of splints shall be such as to avoid laps in penetration test area. When the wrapping has been completed, the movable platform shall be pulled out from under the test specimen forming approximately a one inch by 6-1/2 inch slot test area under the specimen. The modified Vicat plunger containing a size 4 sharp needle snall be placed over the slot test area on the surface of the test specimen and allowed to penetrate. The exposed length of the needle shall be 3/4 inch minimum in order to completely penetrate the five layers of the specimen (see figure 2 for plunger and needle apparatus). The penetration shall be observed with the two mirrors in the apparatus. The first needle penetration shall start about 1/2 inch from the left hand edge of the opening and successive penetrations shall be made from this point to the right at 15 second intervals until the needle no longer pierces through the five layer thick specimen. Start testing at 15 second intervals at 1/2 minute before the lower limit of the required setting range. The needle shall always be kept in the middle of the one rinch wide test area. The needle test shall be repeated until three successive penetrations do not pierce through the test specimen. The first of these, that is one-half minute prior to the last penetration, shall be the setting time. The needle shall be used for five setting time determinations and then replaced. Care shall be taken to keep the needle perpendicular to the test specimen during each test and under no circumstance should any positive or negative force be applied to the Vicat plunger. It must be allowed to rest on the surface and penetrate due to its own weight.

4.4.9 Water penetration. Water penetration shall be determined on the same specimen used for setting time (see 4.4.8). The bandage shall be carefully examined for dry spots or areas after the five second water immersion and while wrapping around the setting time apparatus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

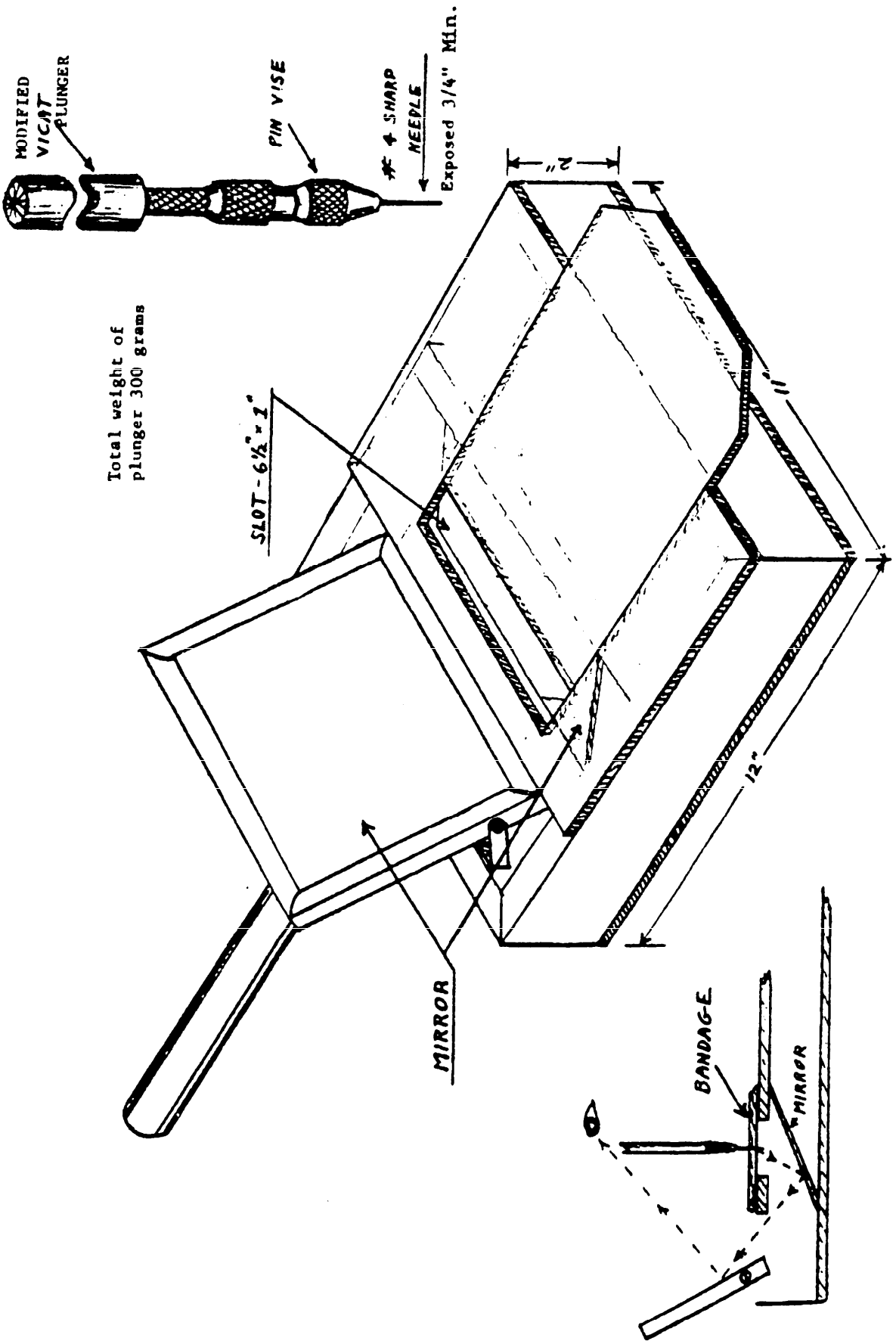


Figure 2. Setting Time Apparatus

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

4.4.10 Leakage test. Prior to filling the bag with bandages or splints, 2000 ± 50 milliliters (ml) of water at 25 ± 1° C shall be poured into the polyethyene bag used to hold the 12 bandages or 50 splints (see 5.1.1.2.1). The opening of the bag shall be loosely closed by hand gathering, and the water-filled bag shall be suspended for 1 minute. During the suspension time, the water-filled bag shall be observed for any leaking.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116 for method IA-8. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Individual bandage preparation - Class A.

5.1.1.1.1 Individual bandage - Type I, class A. Each bandage may be prepared by either of the below specified.

5.1.1.1.1.1 Waxed paper. Each bandage shall be wrapped in waxed paper having a finished weight of not less than 35 pounds (24 inches by 36 inches by 500 sheets). The wax paper shall contain a minimum of 30 percent paraffin, by weight, which has been evenly and smoothly applied to both sides of the basic paper. The length of the waxed paper shall be sufficient to overlap, in the girth direction of the bandage, for a distance of not less than 25 percent. The width of the waxed paper shall be such that the ends protruding over the edges of the bandages, when properly folded, shall completely enclose the bandage in the waxed paper. The finished wrap shall be such that under normal handling the wrapping will not open, or in any way expose the enclosed bandage. The manufacturer may place his name or trademark on the wrapper.

5.1.1.1.1.2 Polyethylene coated paper. Each bandage shall be wrapped in polyethylene coated paper, having a finished weight of not less than 31 pounds (24 in. by 36 in. by 500 sheets). The polyethylene coated paper shall contain a minimum of 28 percent polyethylene by weight, which has been evenly and smoothly applied to one side of the basic paper. The polyethylene coated side shall constitute the inner surface of the wrapper. The length of the polyethylene coated wrapper shall be heat sealed in the girth direction of the bandage. Both ends of the wrapper shall also be heat sealed. The finished wrap of polyethylene coated paper shall be such that under normal handling the wrapping will not open, or in any way expose the enclosed bandage. The manufacturer may place his name or trademark on the wrapper.

5.1.1.1.1.3 Polyethylene bag. Each bandage shall be sealed in a high density linear polyethylene bag of 1.5 ± .05 ml thickness. Polyethylene shall have a moisture vapor transmission rate of not more than 0.48 gm in 24 hours per 100 square in.

5.1.1.1.2 Individual bandage - Type II, Class A. Each bandage may be prepared by either of the below specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.1.1.1.2.1 Foil laminated paper. Each bandage shall be wrapped in a foil laminated paper having a finished weight of not less than 45 pounds (24 inches by 36 inches by 500 sheets). The paper shall be laminated to a minimum .0003 inch thick aluminum foil with wax blend. The surface of the paper shall be uniformly coated with a minimum 6 pound (24 inches by 36 inches by 500 sheets) of wax blend. The length of the wrapper and both ends shall be heat sealed. The finished wrap shall not open in any way, exposing the enclosed bandage, under normal handling. The manufacturer may place his name or trademark on the wrapper. As an alternate, each bandage shall be wrapped in a foil laminated paper having a finished weight of not less than 45 pounds (24 inches by 36 inches by 500 sheets). The paper shall be laminated to a minimum of 0 .000285 inch thick foil with not less than 4.5 pounds of virgin, low density, polyethylene. The inner part of the foil/polyethylene shall be uniformly coated with a minimum of 4.5 pounds of virgin, low density polyethylene. The length of the wrapper and both ends shall be heat sealed. The finished wrap shall not come off in any way, exposing the enclosed bandage. The manufacturer may place his name or trademark on the wrapper.

5.1.1.1.2.2 Polyethylene coated paper. Each bandage shall be wrapped in polyethylene coated paper, having a finished weight of not less than 31 pounds (24 inches by 36 inches by 500 sheets). The polyethylene coated paper shall contain a minimum of 28 percent polyethylene by weight, which has been evenly and smoothly applied to one side of the basic paper. The polyethylene side shall constitute the inner surface of the wrapper. The length of the polyethylene coated wrapper shall be heat sealed in the girth direction of the bandage. Both ends of the wrapper shall also be heat sealed. The finished wrap of the polyethylene coated paper shall be such that under normal handling the wrapping will not open, or in any way expose the enclosed bandage. The manufacturer may place his name or trademark on the wrapper.

5.1.1.1.3 Polyethylene Bag - Type II, Class A. Each type II, Class A bandage wrapped as specified shall be as further packaged in a sealed high density linear polyethylene bag of 1.5 $\pm$ .05 ml thickness. The polyethylene shall have a moisture vapor transmission rate of not more than 0.48 gm in 24 hours per 100 square inch.

5.1.1.3 Unit package (Unit container for civil agencies).

5.1.1.3.1 Bandages. Unless otherwise specified, twelve bandages or fifty splints prepared as specified, (5.1.1.1 and 5.1.1.2) shall be further packaged in a box constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the box specification. As an alternate, a tuck top style fiberboard box may be used. The fiberboard used may be E flute and shall have a minimum bursting strength of 175 lbs. p.s.i. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.1.2 Splint preparation, Type I and II, Class B.

5.1.1.2.1 Polyethylene bag. Fifty splints of one type, class, and size shall be evenly stacked and packaged in a bag fabricated from polyethylene, seamless tubing having an average wall thickness of not less than 0.004 inch, with no wall thickness at any point below 0.0035 inch. Excess air shall be expelled and closure effected by heat sealing. The bag shall be suitable for use as a water receptacle for immersing bandages. The bag shall show no water-leakage when tested as specified in 4.4.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The level C preservation shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. The number of units specified in table VII shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the box specification.

5.2.2 Level B. The number of units specified in table VII shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the box specification.

TABLE VII. Packaging Quantities.

Type	Class	Size	Number of unit packages to be overpacked in each exterior container
I & II	A	2 inches by 3 yards	6
I & II	A	3 inches by 3 yards	6
I & II	A	4 inches by 5 yards	6
I & II	A	5 inches by 5 yards	4
I & II	A	6 inches by 5 yards	4
I	A	8 inches by 5 yards	4
I & II	B	3 inches by 15 inches	12
I & II	B	4 inches by 15 inches	12
I & II	B	5 inches by 30 inches	6
I	B	5 inches by 45 inches	4

5.2.3 Level C. The level C packing shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads commensurate with the level of packing specified in the contract or purchase order, shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Bandages or splints packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

5.2.4.2 Level C. Bandages or splints packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Civil agencies. Marking shall be as specified in FED-STD-123 and shall include any special markings specified in the contract or order.

5.3.2 Military.

5.3.2.1 Individual bandage, Types I and II, Class A. Each bandage shall be commercially identified on the wrap or bag.

5.3.2.2 Unit package. Each package shall be marked as specified in MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act. Type II shelf life markings as specified in MIL-STD-129 shall also be shown. Marking shall also include the legend, "Store at controlled room temperature (59°-86°F)" and the lot number.

Note: In a continuous manufacuring process, the date of manufacture shall be considered the lot or control number.

5.3.2.3 Exterior container and unitized loads. Exterior containers and unitized loads shall be marked as specified in MIL-STD-129. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall also include the legend, "Store at controlled room temperature (59°-86°F)" and the lot number.

5.4 General

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T11:48Z
Check the source to verify that this is the currentversion before use.

6.2 National Stock Number Coverage. This specification covers the following items appearing in the Federal Supply Catalog.

Stock Number	Item Identification
6510-00-201-2001	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Fast Setting, 3 inches by 3 yards.
6510-00-597-2361	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Fast Setting, 4 inches by 5 yards.
6510-00-926-9166	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Extra-Fast Setting, 2 inches by 3 yards.
6510-00-926-9167	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Extra-Fast Setting, 3 inches by 3 yards.
6510-00-926-9168	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Extra-Fast Setting, 4 inches by 5 yards.
6510-00-926-9169	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Extra-Fast Setting, 6 inches by 5 yards.
6510-01-036-9064	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Fast Setting, 6 inches by 5 yards.
6510-01-038-7709	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Fast Setting, 4 inches by 15 inches.
6510-01-038-7710	BANDAGE, CAST, PLASTER OF PARIS IMPREGNATED, Fast Setting, 5 inches by 30 inches.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review Activities:
Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS

PREPARING ACTIVITY:

DoD - MB

AGENT:

DLA - DM

Project No. 6510-0715

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:48Z
Check the source to verify that this is the current version before use.